From Flatcar development to ProxmoxVE deployment

An example of open and sustainable collaboration

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Bonjour 👋

Alexandre

Mathieu

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The takeaways

- Introduction to immutable OS
- Open-source collaboration
- Flatcar deployment on ProxmoxVE

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Did you say “Flatcar deployment on Proxmox”?

Demo is on YouTube: https://youtu.be/S52skKz3LLU

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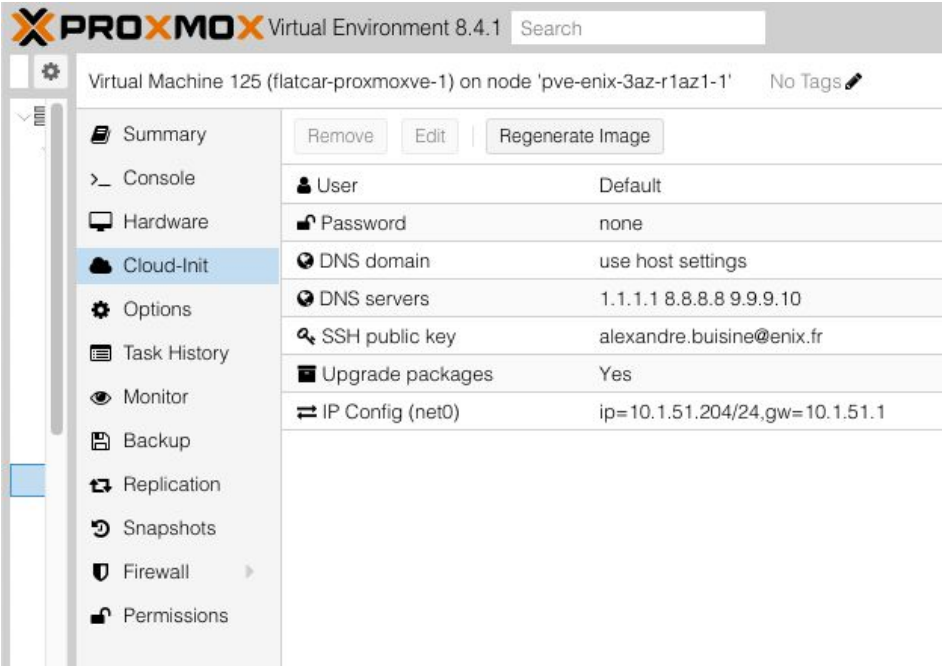
Ignition ou cloud config ?



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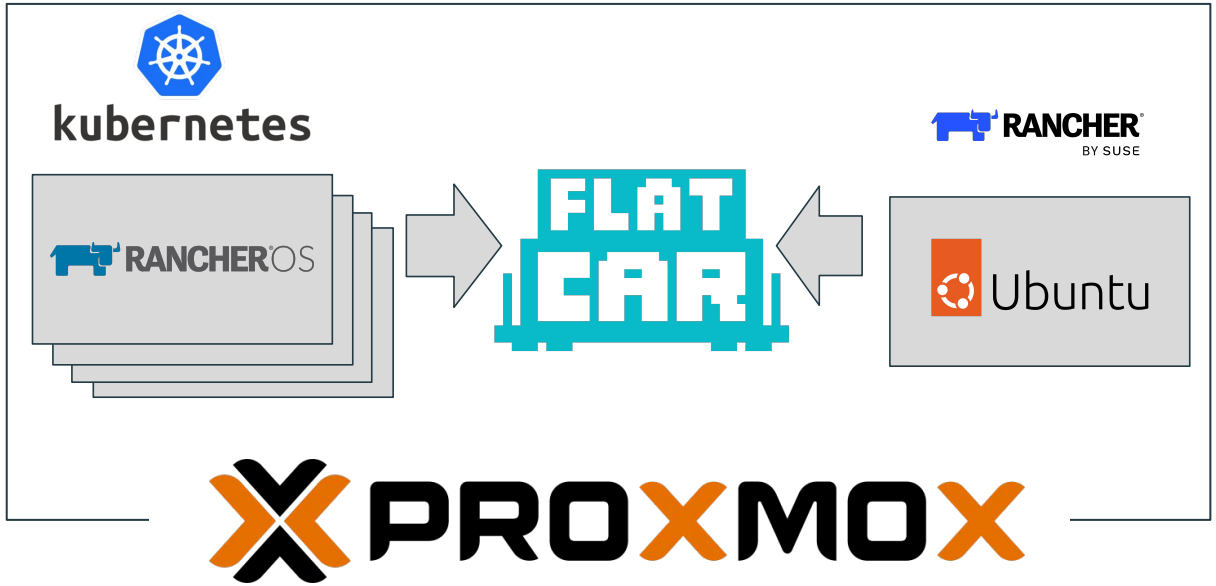
In practice



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Flatcar on ProxmoxVE
-
An open-source journey

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It all starts with an issue …

Proxmox support kind/feature platform/new-platform

#1040 · by lukasmrtvy was closed on Nov 13, 2024

May 2023

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… and ends with a release

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#1040 · by lukasmrtvy was closed on Nov 13, 2024

May 2023

November 2024

Flatcar Container Linux 🚂
@flatcar@hachyderm.io

New Flatcar Alpha, Beta and Stable major releases now available!
📦 Many package updates: Docker 26, Curl, Podman sysext
🚀 New features: Proxmox support in Alpha
🔒 CVE fixes & security patches: Docker, Kernel, Curl
📜 Release notes at the usual spot: flatcar.org/releases/

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Adding a new platform

Proxmox support kind/feature platform/new-platform

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?

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Adding a new platform

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May 2023

- Afterburn support — June 2024
- Ignition support — August 2024
- OEM support — November 2024
- (Test implementation) — November 2024

November 2024

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@flatcar@hachyderm.io

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- Documentation — June 2025

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Adding a new platform

- Ignition support
- Afterburn support
- OEM support
- Documentation
- (Test implementation)

Fedora CoreOS

Flatcar

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- **Ignition support**
- Afterburn support
- OEM support
- Documentation
- (Test implementation)

Ignition support

- One time provisioning software
- Used by FCOS, Flatcar, MicroOS, …
- Run from the initramfs
- Single binary with some system dependencies
- Provision or nothing

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Ignition support

- **Ignition support**
- Afterburn support
- OEM support
- Documentation
- (Test implementation)

```yaml
variant: fcos
version: 1.1.0
storage:
  files:
    - path: /opt/file
      contents:
        inline: Hello, world!
      mode: 0644
      user:
        id: 500
      group:
        id: 501
```

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Ignition support

- **Ignition support**
- Afterburn support
- OEM support
- Documentation
- (Test implementation)

providers: support for Proxmox VE (continuation) #1910

Merged **jlebon** merged 1 commit into `coreos:main` from `enix:proxmoxve` on Aug 6, 2024

Conversation 12 | Commits 1 | Checks 0 | Files changed 4

bitdriftr commented on Jul 17, 2024 — Contributor

Hello, this PR is a continuation of #1790. The original PR is doing good, but we cannot push to it and we would like things to go faster so we decided to make another one to finish it.

This PR includes the commit from @b-, and a fix for #1790 (comment). I also added release note.

I think this is good to go. Feel free to tell me otherwise.

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- Ignition support
- **Afterburn support**
- OEM support
- Documentation
- (Test implementation)

Afterburn support

- Afterburn runs *after* the boot
- Help to:
  - Fetch SSH keys from the instance metadata
  - Fetch the instance metadata (e.g IP, hostname, instance ID, etc.)
  - Configure the hostname

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Afterburn support

- Ignition support
- **Afterburn support**
- OEM support
- Documentation
- (Test implementation)

providers: support for proxmoxve #1023

Merged **jlebon** merged 1 commit into `coreos:main` from `enix:proxmoxve` on Jun 19, 2024

Conversation 33 | Commits 1 | Checks 0 | Files changed 27

bitdriftr commented on Dec 19, 2023 — Contributor

Hello,

This PR adds support for reading the cloud init drive from a ProxmoxVE instance.
The code mounts `/dev/disk/by-label/cidata` to a temprary location and then parses the YAML files present inside.
It supports : setting hostname, writing ssh keys, and writing systemd network configuration.

The changes introduces a new OEM ID `proxmoxve` that will be matched from the kernel command line. As time of writing, this OEM ID is not yet recognized by Ignition or other tools.

We currently maintain a build script that modifies a Flatcar image to inject this version of Afterburn : https://github.com/enix/flatcar-proxmoxve

Feedback is welcome :)

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OEM Support
(a.k.a “Flatcar Proxmox images”)

- Ignition support
- Afterburn support
- **OEM support**
- Documentation
- (Test implementation)

OEM: Provide Proxmox images #1783

Merged **tormath1** merged 2 commits into `main` from `kai/proxmox-support` on Nov 12, 2024

Conversation 54 | Commits 2 | Checks 5 | Files changed 13

pothos commented on Mar 22, 2024 • edited by tormath1 Member

This pulls in flatcar/bootengine#91
and flatcar/init#115 to run afterburn for
hostname, network, SSH key, and metadata attribute setup. The afterburn
support for the SSH key and hostname parses the user-data when it's
cloud-init. The coreos-cloudinit support is not there but can be added
in addition: We need to add a new provider that varies from the existing
config drive support because the file is called user-data and not
user_data, and it needs to look for a filesystem label cidata and not
config-2.

Related to: flatcar/Flatcar#1040

Documentation

- Ignition support
- Afterburn support
- OEM support
- **Documentation**
- (Test implementation)

0 Open ✓ **4 Closed** Auth

proxmox: move the documentation to 'community-platforms' ✓
#446 by tormath1 was merged on Jun 27 • Approved

docs: proxmoxve: add promoxve install doc
#445 by fhemberger was merged on Jun 27 • Review required

kubernetes/capi: add kubevirt and proxmox ✓
#402 by tormath1 was merged on Mar 25 • Approved

docs: proxmoxve: add promoxve install doc
#337 by bitdriftr was closed on Jun 27 • Draft

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Documentation

- Ignition support
- Afterburn support
- OEM support
- **Documentation**
- (Test implementation)

- https://www.flatcar.org/docs/latest/installing/community-platforms/proxmoxve/

Documentation > latest > Installing > Community supported platforms > Running on Proxmox VE

Running Flatcar Container Linux on Proxmox VE

Please note that Proxmox VE is not an officially supported platform at this time because the release tests don't run for it.

These instructions will walk you through running Flatcar Container Linux on Proxmox VE.

Choosing a channel

Flatcar Container Linux is designed to be updated automatically with different schedules per channel. You can disable this feature , although we don't recommend it. Read the release notes for specific features and bug fixes.

Stable Channel Beta Channel Alpha Channel

The Stable channel should be used by production clusters. Versions of Flatcar Container Linux are battle-tested within the Beta and Alpha channels before being promoted. The current version is Flatcar Container Linux 4230.2.3.

- Ignition support
- Afterburn support
- OEM support
- Documentation
- **(Test implementation)**

Test

- About the Flatcar tests

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Test

- Ignition support
- Afterburn support
- OEM support
- Documentation
- **(Test implementation)**

Proxmox testing with community #1573

Closed **tormath1** started this conversation in **Show and tell**

tormath1 on Oct 25, 2024 Maintainer

A Proxmox image is available here for testing purposes here: http://bincache.flatcar-linux.net/images/amd64/9999.9.100+kai-proxmox-support/flatcar_production_proxmoxve_image.img.bz2.

One can try it following this documentation: https://github.com/flatcar/flatcar-website/pull/337/files or Terraform approach. You can directly suggest your feedback to the documentation or discuss about it in this discussion:

1

3 comments · 17 replies

Oldest Newest Top

ader1990 on Oct 25, 2024 Maintainer

I could succesfully create and run a Flatcar VM with Ignition on Proxmox VE latest (8.2.2):

```
export VM_ID=124
```

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So, did you say “Flatcar deployment on Proxmox”?

Demo is on YouTube: https://youtu.be/bZY2b7eKyTY

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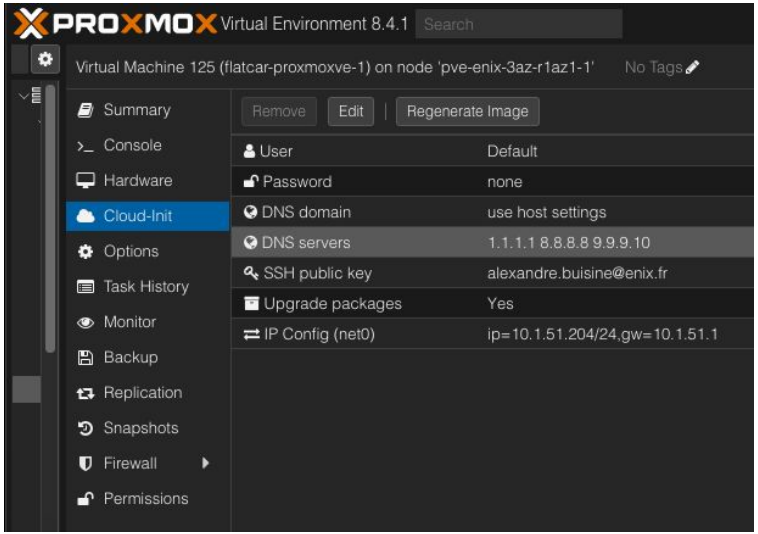
Résultat sur la machine via cloud config




```
flatcar-proxmoxve-1 ~ # cat /run/metadata/flatcar
COREOS_PROXMOXVE_INSTANCE_ID=c26d5b4c230c74156c49a1a533243cf0f6e78f66
COREOS_PROXMOXVE_IPV4=10.1.51.204
COREOS_PROXMOXVE_HOSTNAME=flatcar-proxmoxve-1
flatcar-proxmoxve-1 ~ #
flatcar-proxmoxve-1 ~ #
flatcar-proxmoxve-1 ~ # networkctl status
● Interfaces: 23, 3, 2, 1
       State: routable
Online state: online
     Address: 10.1.51.204 on eth0
              172.17.0.1 on docker0
              fe80::be24:11ff:feb1:e92a on eth0
              fe80::42:b1ff:fed2:ece0 on docker0
              fe80::c810:d5ff:fe71:ca59 on veth330af44
     Gateway: 10.1.51.1 on eth0
         DNS: 1.1.1.1
              8.8.8.8
              9.9.9.10
```

Result of this contribution

- Fedora CoreOS, MicroOS to come on Proxmox? (e.g: https://github.com/coreos/fedora-coreos-tracker/issues/736)
- Unlocked Flatcar support on Cluster API Proxmox
- Terraform/OpenTofu examples to provide
- Contribute to sustainable collaboration between communities

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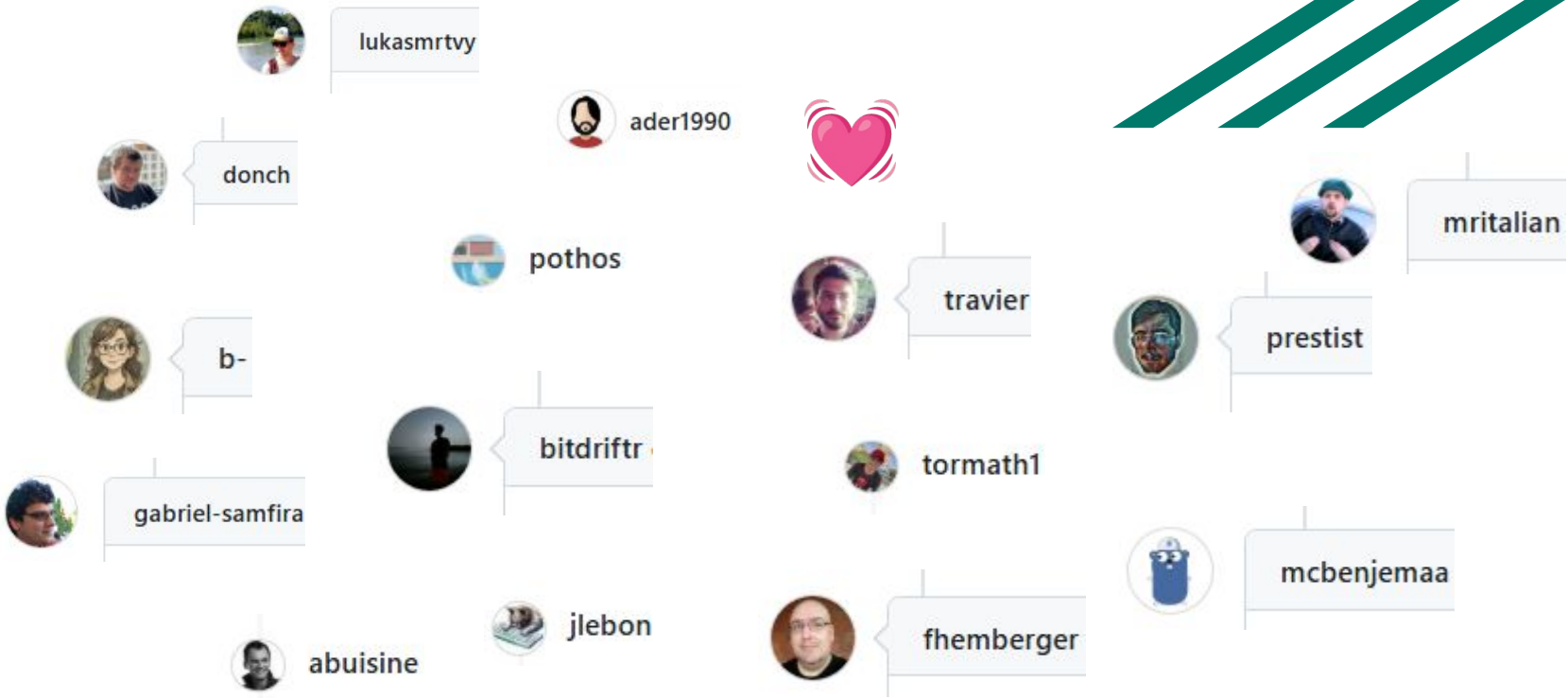

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Merci pour votre écoute

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